

Soybean Price ECM-GARCH Model

By kittipmn25@gmail.com | <https://github.com/KittiP1150/Soybean-Price-ECM-GARCH-Model>

1.Data

The data is pull from yahoo finance

soybean (ZS=F)

soybean meal (ZM=F)

usd_thb (THB=X)

2.Methodology

The development of the trading strategy for Soybean futures contracts was conducted within the framework of financial econometrics. The proposed methodology was founded upon the evolution of time-series analysis and was structured into three sequential stages, as follows

2.1 Wold Representation Theorem

The theorem states that any covariance-stationary time series can be decomposed into two components

$$X_t = \sum_{j=0}^{\infty} d_j \varepsilon_{t-j} + \eta_t \quad (1)$$

The first component, $\sum_{j=0}^{\infty} d_j \varepsilon_{t-j}$ is square summable moving average term, d_j is coefficient, and ε_{t-j} is known as the 'innovation' in X_t . The second component, η_t is perfectly predictable based on past observations. The two parts of this representation are called the 'purely indeterministic' and the 'deterministic' parts

For the ARMA is a special case in Wold's theorem. The ARMA represent

$$y_t = \phi_t y_{t-1} + \dots + \phi_p y_{t-p} + v_t + \theta_1 v_{t-1} + \dots + \theta_q v_{t-q} \quad (2)$$

Where v_t is an Independent and Identically Distributed process which is The each of variables are independent and the distribute pattern are the same. ε_t 's representation as corresponding to the amount of 'memory' in the process. Thus, the ARMA(p,q) models have 'short memory' relative to the entire class representations

While the ARMA representation perfectly models the short-term memory of stationary univariate series, raw financial asset prices are fundamentally non-stationary. Therefore, the mathematical principles of Wold's decomposition and stationarity discussed herein serve as the crucial theoretical foundation for the subsequent Cointegration framework (Section 2.2), which mathematically seeks to extract a stationary equilibrium error $I(0)$ from non-stationary multivariate series.

2.2 Cointegration

Financial asset prices typically exhibit non-stationary $I(1)$ behavior. However, Engle and Granger's (1987) Cointegration theory states that a linear combination of two $I(1)$ series sharing a common stochastic trend can be covariance-stationary $I(0)$, forming a mean-reverting equilibrium. To establish this framework, Ordinary Least Squares (OLS) regression is employed to estimate the optimal Hedge Ratio (γ). The extracted residuals mathematically represent the dynamic price spread deviating from this long-run equilibrium.

From Granger Representation .Given a series with no deterministic component which has a stationary, invertible, ARMA representation after differencing d times, is said to be integrated of order d , denoted $x_t \sim I(d)$.

From

$$z_t = x_t - ay_t$$

will also be $I(d)$, if x_t and y_t are $I(d)$. However, There are special case .We can find " a " that make $z_t \sim I(0)$. In this case " a " is also called Hedge ratio(β)

The optimal Hedge Ratio is estimated via OLS regression with an intercept; the extracted residuals inherently satisfy the first-order condition of least squares. Consequently, the sum of the residuals is strictly zero ($\sum \epsilon_t = 0$), dictating that the resulting price spread strictly oscillates around a zero-mean expected value ($E[\epsilon_t] = \mu = 0$).

Then, we can define trading signals by analyzing the Co-Integration and Error Correction. Granger and Engle present that a vector time series x_t has an error correction representation if it can be expressed as

$$A(B)(1 - B)x_t = -\gamma z_{t-1} + u_t$$

where

$$z_{t-1} = x_{t-1} - \alpha'y_{t-1} \text{ . and } \gamma \neq 0$$

The equation shown that $(1 - B)x_t$ is depend on a stationary multivariate disturbance (u_t) and equilibrium error(z_{t-1}) which is multiplied by the speed of adjustment coefficient (γ). Whenever the equilibrium error (z_{t-1}) widens, this negative coefficient ensures that the subsequent price change $(1 - B)x_t$ will adjust in the opposite direction to correct the divergence.

This continuous pull-back mechanism is statistically known as Mean Reversion. Consequently, it forms the core logic for our trading signals: by identifying extreme values of the spread, the system can systematically open positions with the **expectation that the ECM mechanism will revert the price back to its historical equilibrium.**

2.3 Handling Volatility Clustering with the GARCH(1,1) Model

While the cointegration process can successfully generate a statistically stationary ($I(0)$) price spread, generating trading signals using traditional standard deviation inherently relies on the assumption of constant variance (homoskedasticity).

This assumption contradicts the actual behavior of financial asset prices. Under the traditional Efficient Market Hypothesis, Let $Z(t)$ denote the price process. The equation,

$$L(t, 1) = \ln Z(t + 1) - \ln Z(t)$$

is represent the sequential logarithmic returns

Mandelbrot (1963) provided an analytical solution to model this dependency by applying an affine transformation (coordinate rotation) to the return series. By rotating the coordinate system, the dependency is mapped into two new variables representing the first and second finite differences:

$$S(t) = \frac{1}{2} \times [L(t, 1) + L(t - 1, 1)]$$

and

$$D(t) = \frac{1}{2} \times [L(t, 1) - L(t - 1, 1)]$$

From the independence of $S(t)$ and $D(t)$, the conditional density function of the current return, given a known prior realization $L(t - 1, 1)$, asymptotically follows a Paretian distribution governed by a transformed exponent:

$$P(L(t, 1) > x \mid L(t - 1, 1)) \sim x^{-(2\alpha+1)}$$

This makes the conditional variance become strictly finite and measurable and confirms that the financial market variance is a deterministic function that depends on past states.

Due to the computational intractability of performing likelihood estimation on the moments of a Paretian distribution with a $2\alpha + 1$ exponent, Engle (1982) strategically reformulated Mandelbrot's conceptual framework of $E[|L_t| \mid L_{t-1}]$ into a conditional second

moment. By hypothesizing a linear structural dependency on past squared observations, the conditional variance is expressed as:

$$Var(L_t | \mathcal{F}_{t-1}) = \alpha_0 + \sum_{i=1}^q \alpha_i L_{t-i}^2 \quad (3)$$

Where

$$\mathcal{F}_{t-1} = \{L_{t-1}, L_{t-2}, \dots, L_0\}$$

This mathematical formulation established the Autoregressive Conditional Heteroskedasticity (ARCH) model. Crucially, this linear structure fully inherits the underlying mathematical properties of the Markovian process, thereby perfectly capturing and replicating the empirical phenomenon of volatility clustering.

From

$$L(t) = u_t + \epsilon_t$$

Where u_t represents the conditional expected return, which is the predictable component based on historical data. When analyzing high-frequency financial data, such as daily or intraday intervals, the expected return u_t is statistically negligible compared to the magnitude of the variance. Therefore,

$$u_t = E[L_t | \mathcal{F}_{t-1}] \approx 0$$

From equation 3. ARCH model implement conditional variance as

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \alpha_2 \epsilon_{t-2}^2 + \dots + \alpha_q \epsilon_{t-q}^2$$

Because the standard ARCH model requires an extensively large number of α parameters to capture the long-memory effects of financial markets, it inherently suffers from parameter explosion. To resolve this limitation, the model was advanced by introducing the lagged conditional variance term, $\beta_1 \sigma_{t-1}^2$, establishing the GARCH(1,1) equation:

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

This formulation is subject to the stability conditions $\alpha_1 + \beta_1 < 1$ and $\beta_1 > 0$. This model is revealed through recursive substitution, which mathematically transforms the equation into:

$$\sigma_t^2 = \frac{\omega}{1-\beta_1} + \alpha_1 \sum_{i=1}^{\infty} \epsilon_{t-i}^2 + \beta_1^{i-1} \epsilon_{t-i}^2$$

This derivation proves that the GARCH(1,1) model can successfully incorporate an infinite history of past volatility data into its estimations while strictly requiring only three parameters to be estimated: $(\omega, \alpha_1, \beta_1)$.

2.4 ECM-GARCH

The final stage of the methodology synthesizes the long-run equilibrium properties of the Cointegration model with the short-term volatility dynamics of the GARCH(1,1) framework. The theoretical justification for this integration lies in redefining the distributional properties of the error term.

In a standard Error Correction Model (ECM), the residual ϵ_t , which represents the equilibrium spread, is fundamentally assumed to be homoskedastic white noise. However, having established the presence of volatility clustering (as modeled in Section 2.3), the strict assumption of a constant variance σ^2 is violated. To integrate the GARCH framework, the unconditional distribution of the error term is replaced with a conditional distribution based on the historical information set $\mathcal{F}_{t-1}$. Where σ_t^2 follows the GARCH(1,1) process. In time-series econometrics, an error term exhibiting conditional heteroskedasticity is mathematically decomposed into two multiplicative components:

$$\epsilon_t = z_t \sigma_t$$

Where

ϵ_t is the mean-adjusted divergence from the equilibrium ($spread_t - \mu$)

σ_t is the conditional standard deviation at time t derived from the GARCH model.

z_t is the Standardized Residual, which represents an independent and identically distributed (i.i.d.) random variable with a mean of zero and a variance of one

By algebraically rearranging the structural decomposition equation, the standardized residual z_t is isolated:

$$z_t = \frac{\epsilon_t}{\sigma_t}$$

the trading signal is formally derived as a Risk-Adjusted Dynamic Z-Score:

$$Z_t = \frac{spread_t - \mu}{\sigma_t}$$

During periods of market tranquility, the threshold narrows, preserving signal sensitivity for short-term mean-reversion. Conversely, during periods of heightened volatility clustering, the inflation of the denominator σ_{GARCH} automatically widens the threshold. This mathematical self-preservation mechanism prevents premature market entries

2.5 Backtest

To validate the profitability and assess the risk profile of the Pairs Trading strategy utilizing the ECM-GARCH model, this study constructs a comprehensive backtesting system. The system simulates a virtual trading environment using historical data. The backtesting framework is strictly designed to prevent any look-ahead bias, ensuring the robustness and validity of the empirical results. The evaluation components are detailed as follows:

2.5.1 Trading Logic and State Transitions

The decision to enter or exit a trading position is governed by the risk-adjusted dynamic Z-score (Z_t), synthetically derived in Section 2.4. The operational thresholds are defined as

1) Entry Signals:

- Long Spread: The system initiates a long position in the primary asset and a short position in the secondary asset, weighted by the optimal hedge ratio, when $Z_t < -2.0$. This indicates that the price spread is significantly undervalued relative to its historical equilibrium.
- Short Spread: The system initiates a short position in the primary asset and a long position in the secondary asset when $Z_t > 2.0$, indicating that the spread is significantly overvalued.

2) Exit Signals (Mean Reversion):

- The system clears all active positions to realize profit or loss when Z_t reverts and crosses the zero line ($Z_t \approx 0$). This zero-cross signifies that the spread has fully reverted to its econometric equilibrium.

2.5.2 Strategy Return Calculation

During the simulation, the daily strategy return ($R_{Strategy,t}$) is calculated by multiplying the spread return by the direction of the held position (trade signal), mathematically expressed as:+

$$R_{Strategy,t} = R_{y,t} - (Hedge\ ratio \times R_{x,t})$$

$$R_{Strategy,t} = Signal_{t-1} \times R_{spread,t}$$

Where

$R_{y,t}$ and $R_{x,t}$ represent the daily returns of the primary and secondary assets.

Hedge ratio is the optimal hedge ratio estimated via the Ordinary Least Squares (OLS) model.

$Signal_{t-1}$ denotes the trading signal generated at the end of the previous day (1 for Long, -1 for Short, and 0 for Neutral position). Utilizing the lagged signal strictly enforces the no look-ahead bias constraint, ensuring trades are executed based solely on prior available information.

2.5.3 Performance and Risk Metrics

To evaluate the model's robustness against institutional quantitative standards, this study employs four primary metrics

1. Cumulative Total Return Measures the overall portfolio growth from the inception to the end of the testing period

$$Total\ Return = \prod_{t=1}^T (1 + R_{Strategy,t}) - 1$$

2. Annualized Return Standardizes the total return into a yearly rate, facilitating comparison with risk-free rates or other asset classes

$$Annualized\ Return = (1 + Total\ Return)^{\frac{252}{N}} - 1$$

3. Maximum Drawdown Considered the most critical risk metric, it represents the largest peak-to-trough drop in the portfolio's equity value. It indicates the worst-case scenario an investor might endure during the investment horizon

$$Drawdown = \frac{Equity_t - \max(Equity_1, \dots, Equity_t)}{\max(Equity_1, \dots, Equity_t)}$$

4. Sharpe Ratio Measures the risk-adjusted return. A Sharpe ratio exceeding 1.0 indicates that the strategy generates excellent returns relative to the volatility assumed

$$Sharpe\ Ratio = \frac{Annualized\ Return - Risk\ Free\ Rate}{Annualized\ Volatility}$$

3.Design

3.1 System Architecture Overview

The pairs trading system utilizing the ECM-GARCH model is constructed upon a 3-Tier Architecture, comprising the data layer, the core logic layer, and the display layer. The architectural workflow is illustrated in Figure 3.1.

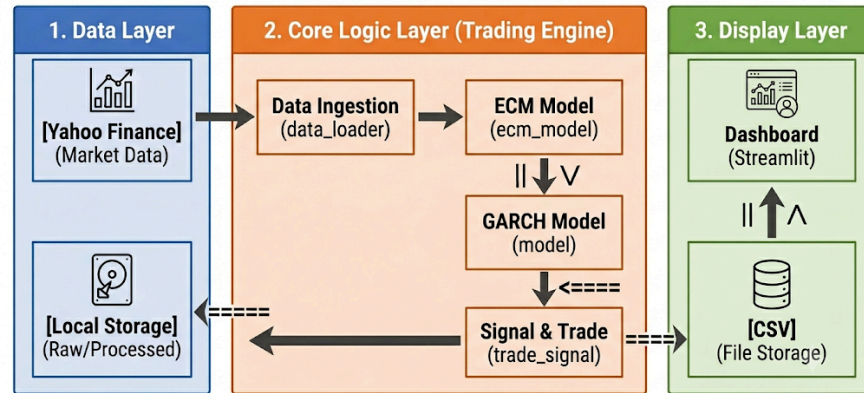


Figure 3.1 architectural workflow

The fundamental components of the system are categorized as

1. Data Layer: Responsible for retrieving raw asset price data from external APIs (e.g., Yahoo Finance) and securely storing it within a local file system. The storage is strictly partitioned into raw data (raw) and cleaned data (processed) directories to preserve data integrity.
2. Core Logic Layer: Serving as the central processing unit (Trading Engine) developed in Python. This layer is responsible for extracting the cointegration price spread, estimating conditional volatility, and generating execution signals based on established econometric principles.
3. Display Layer: Designed to receive the final output data from the processing layer and present it via an interactive graphical interface (Dashboard). This empowers users or portfolio managers to visually monitor the equity curve and assess portfolio risk metrics in a tangible format.

To prevent code entanglement and ensure long-term maintainability, this research structures the software into functional modules utilizing a Modular Design framework. The data pipeline flows sequentially through the following modular components:

1. `data_loader.py` (Data Ingestion & Cleaning): Imports historical price data (Open, High, Low, Close) for soybean meal and soybeans. It executes data wrangling processes, including dropping missing values (NaN) and normalizing the price scale to a unified baseline.

2. `ecm_model.py` (Cointegration Engine): Processes the cleaned data through an Ordinary Least Squares (OLS) regression. This module estimates the optimal hedge ratio and extracts the residuals, representing the stationary price spread (Error Term).
3. `model.py` (Volatility Estimation): Ingests the stationary spread into a GARCH(1,1) mathematical model. By strictly imposing a zero-mean constraint, it evaluates the current conditional volatility of the spread.
4. `trade_signal.py` (Decision Engine): Computes the dynamic Z-Score by standardizing the spread against the conditional volatility. If the Z-Score breaches the predefined operational threshold, the system triggers and logs the corresponding trade execution state.
5. `backtest.py` (Portfolio Simulation): Integrates the generated signals with historical pricing data to simulate real-world portfolio execution. It calculates the daily strategy return, constructs the cumulative equity curve, and evaluates institutional-grade risk metrics, including Maximum Drawdown and the Sharpe Ratio.

The entirety of this pipeline is orchestrated by a central controller script (`main.py`). This script governs the sequential execution of all underlying modules and exports the final consolidated results to a CSV file (or a Time-Series Database) for subsequent rendering on the display layer.

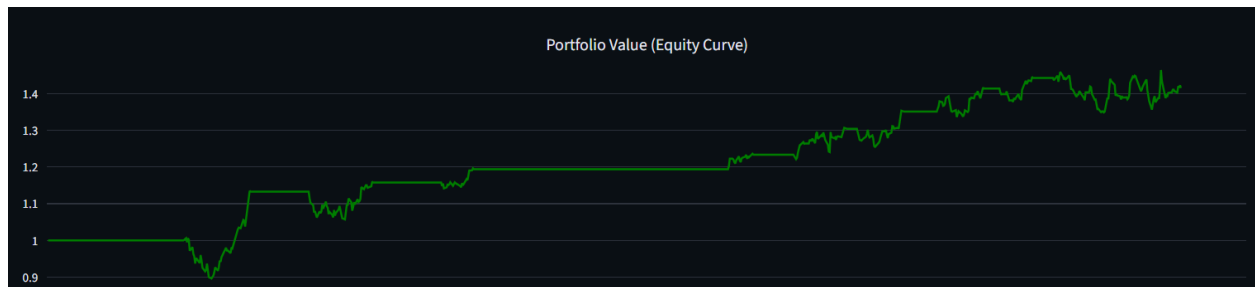
4.Result

Table 4.1 presents the results of the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) unit root tests for the Soybean and Soybean Meal future prices. The results indicate that both price series are non-stationary in their levels ($I(1)$), failing to reject the null hypothesis of the ADF test while rejecting the null hypothesis of the KPSS test. However, after applying first-order differencing (Δ), both series exhibit stationarity ($I(0)$), satisfying the prerequisite for subsequent Cointegration and Error Correction Model (ECM) analysis.

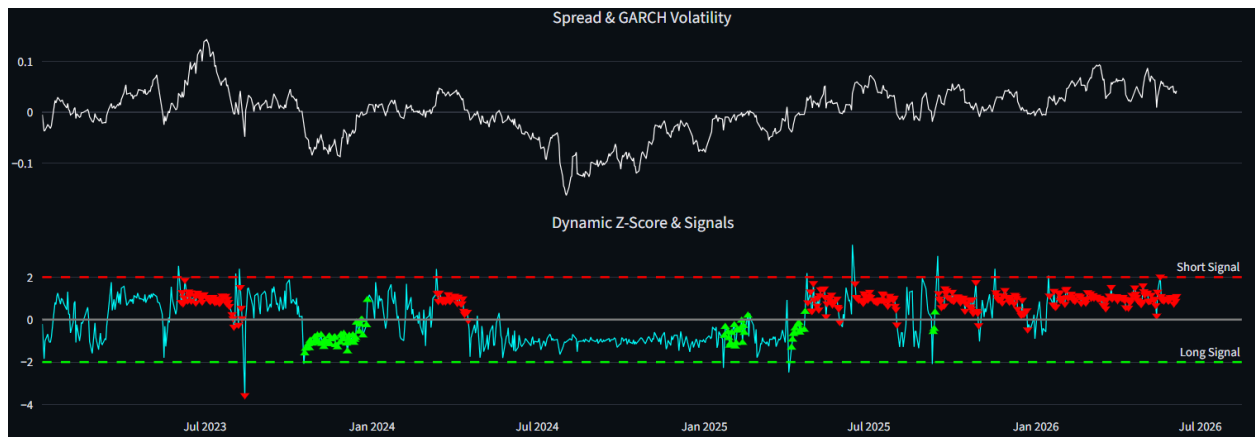
Series	ADF p-value	KPSS p-value	Conclusion
Soybean Future Price	0.2435	0.010	Non-stationary
Soybean Meal Future Price	0.2560	0.010	Non-stationary
Δ Soybean Future Price	0.0000	0.100	Stationary $\rightarrow I(1)$
Δ Soybean Meal Future Price	0.0000	0.100	Stationary $\rightarrow I(1)$

As observed in the early phase of the equity curve, the portfolio experienced an initial decline (corresponding to the Maximum Drawdown of -10.88%). Following this period, the system

successfully recovered and consistently generated new equity highs, ultimately driving the portfolio value to nearly 1.5. This trajectory precisely aligns with the final Total Return of 41.49%. (Assume Transaction fee is 0.1%)



Spread & GARCH Volatility: The spread clearly exhibits mean-reverting behavior, consistently oscillating around the zero line.



Reference

- 1.Wold, H. (1938). *A Study in the Analysis of Stationary Time Series*. Uppsala: Almqvist & Wiksell.
- 2.Engle, R. F., & Granger, C. W. J. (1987). Co-Integration and Error Correction: Representation, Estimation, and Testing. *Econometrica*, 55(2), 251–276. <https://doi.org/10.2307/1913236>
- 3.Vidyamurthy, G. (2004). *Pairs Trading: Quantitative Methods and Analysis*. John Wiley & Sons.
4. Mandelbrot, B. (1963). The Variation of Certain Speculative Prices. *The Journal of Business*, 36(4), 394-419.
- 5.Engle, R. F. (1982). Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. *Econometrica*, 50(4), 987-1007.
- 6.Chen, C. W. S., et al. (2001). Pairs Trading via Three-Regime Threshold Autoregressive GARCH Models.